

# Output Ontology (OO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: WXIMPACTBENCH | Context: Gran Chaco Tri-Border Emergency Response Zone

**Output Ontology definition:** Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

## Justification

The benchmark uses six binary impact labels [Q25, Q117] with detailed Canadian-anchored operational definitions [Q141–Q146] and a multi-label simultaneous classification structure [Q57] that nominally matches the deployment's minimum acceptable output form. However, the boundary logic is misaligned with the consequence-framing-dependent assignment that Gran Chaco national coordinators apply. Dataset analysis reveals visible label boundary inconsistencies even within the Canadian context: 100,000 homes damaged labeled agricultural+health but not infrastructural [D16]; storm with skull-fracture casualty labeled all-zero [D6]; one million people without power labeled all-zero [D9]. The Political category exemplars (municipal snow removal [D10], policy debates [Q140]) lack any border integrity or cross-border displacement framing. The Agricultural category emphasizes Canadian farm-context exemplars [Q136, Q142] without dirt-road access disruption to indigenous communities. Score is 2 (not 1) because the multi-label binary structure itself is reusable and partially aligned with deployment needs.

| ID   | Evidence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| Q25  | "Six vulnerability-related disruptive weather impacts are defined as the labeling categories, including Infrastructural, Political, Financial, Ecological, Agricultural, and Human Health." (p.4)                                                                                                                                                                                                                                                                                                                                                                                              |
| Q57  | "Different from traditional methods that decompose multi-label text classification into multiple binary classification tasks (Boutell et al., 2004; Liu et al., 2017), we simultaneously identify all relevant disruptive weather impacts for each input by calling the LLM once." (p.6)                                                                                                                                                                                                                                                                                                       |
| Q117 | "The classification task is binary (true/false), requiring the model to identify whether the text explicitly mentions any of the defined impacts." (p.16)                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Q136 | "Agricultural Impact: Focuses on weather-related effects on farming and livestock management. Includes crop yield variations; direct damage to crops, timber, or livestock; modifications to farming schedules; disruptions to food production and supply chains; impacts on farming equipment; and changes in agricultural inputs (e.g., soil conditions, water availability, fertilizers, animal feed). Both immediate and long-term effects should be considered." (p.19)                                                                                                                   |
| Q140 | "Political Impact: Evaluates governmental and policy responses to weather events. Includes government decision-making and policy changes; shifts in public opinion or political discourse; effects on electoral processes; international aid and relations; and debates on disaster preparedness and response. Both direct political reactions and policy implications should be analyzed." (p.19)                                                                                                                                                                                             |
| Q141 | "Infrastructural Impact: Classify as 'true' if the text mentions any damage or disruption to physical infrastructure and essential services. This includes structural damage to buildings, roads, or bridges; any disruptions to transportation systems such as railway cancellations or road closures; interruptions to public utilities including power and water supply; any failures in communication networks; or damage to industrial facilities. Consider only explicit mentions of physical damage or service disruptions in your classification." (p.20)                              |
| Q142 | "Agricultural Impact: Classify as 'true' if the text mentions any weather-related effects on farming and livestock management operations. This includes yield variations in crops and animal products; direct damage to crops, timber resources, or livestock; modifications to agricultural practices or schedules; disruptions to food production or supply chains; impacts on farming equipment and resources; or effects on agricultural inputs including soil conditions, water availability for farming, and essential materials such as seedlings, fertilizers, or animal feed." (p.20) |
| Q143 | "Ecological Impact: Classify as 'true' if the text mentions any effects on natural environments and ecosystems. This includes alterations to local environments and biodiversity; impacts on wildlife populations and behavior patterns; effects on non-agricultural plant life and vegetation; modifications to natural habitats including water bodies, forests, and wetlands; changes in hydrological systems such as river levels and lake conditions; or impacts on urban plant life." (p.20)                                                                                             |

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| <b>Q144</b> | “Financial Impact: Classify as ‘true’ if the text explicitly mentions economic consequences of weather events. This includes direct monetary losses; business disruptions or closures requiring financial intervention; market price fluctuations or demand changes for specific goods; impacts on tourism and local economic activities; or insurance claims or economic relief measures. Focus only on explicit mentions of financial losses or fluctuations.” (p.20) |
| <b>Q145</b> | “Human Health Impact: Classify as ‘true’ if the text mentions physical or mental health effects of weather events on populations. This includes direct injuries or fatalities (including cases where zero or more casualties are explicitly mentioned); elevated risks of weather-related or secondary illnesses; mental health consequences such as stress or anxiety; impacts on healthcare service accessibility; or long-term health implications.” (p.20)          |
| <b>Q146</b> | “Political Impact: Classify as ‘true’ if the text mentions governmental and policy responses to weather events. This includes government decision-making and policy modifications in response to events; changes in public opinion or political discourse; effects on electoral processes or outcomes; international relations and aid responses; or debates surrounding disaster preparedness and response capabilities.” (p.20)                                       |
| <b>D6</b>   | A poor lady passenger was dashed to leeward and had her skull fractured. She was landed this morning in a dying state with a coffined child that succumbed last night to its sufferings. — Storm narrative with human casualty but annotated all-zero — potential label discrepancy                                                                                                                                                                                     |
| <b>D9</b>   | residents of southwestern Quebec have been plunged back into the deep freeze Overnight lows in Montreal were expected to drop to minus-16 Celsius last night and tumble as far as minus-30C on the South Shore, where more than one million people struggle on without power or heat — 1998 Quebec ice storm aftermath — one million without power/heat, annotated all-zero                                                                                             |
| <b>D10</b>  | Residents of the locality were put to great inconvenience by the huge piles of snow now accumulated on the streets... Already \$30,000 has been expended in removing it — 1887 Montreal snow removal municipal debate — multi-label (infra+pol+fin) correctly labeled                                                                                                                                                                                                   |
| <b>D16</b>  | More than 292,000 people have been evacuated in the mountainous region along the Yangtze River, with more than 100,000 homes damaged and crops on about 175,000 hectares of farmland destroyed — 2007 European/Chinese/Pakistani multi-location disaster report from Montreal Gazette — agricultural and health labeled, but not infrastructural despite 100k homes damaged                                                                                             |

## Strengths

- Multi-label simultaneous binary structure [Q57, Q117] matches the deployment's minimum acceptable output form
- Each sample may carry multiple labels simultaneously [Q28, D10] — structurally compatible with consequence-framing where one event attracts multiple labels
- Detailed operational definitions exist [Q135–Q146], providing a starting template that could be adapted/expanded for Gran Chaco contexts

| ID          | Evidence                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| <b>Q28</b>  | “Unlike previous study (Imran et al., 2016a), however, each sample might correspond to more than one impact.” (p.4)                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Q57</b>  | “Different from traditional methods that decompose multi-label text classification into multiple binary classification tasks (Boutell et al., 2004; Liu et al., 2017), we simultaneously identify all relevant disruptive weather impacts for each input by calling the LLM once.” (p.6)                                                                                                                                                                                            |
| <b>Q117</b> | “The classification task is binary (true/false), requiring the model to identify whether the text explicitly mentions any of the defined impacts.” (p.16)                                                                                                                                                                                                                                                                                                                           |
| <b>Q135</b> | “Infrastructural Impact: Examines weather-related damage or disruption to physical infrastructure and essential services. Includes structural damage to buildings, roads, and bridges; disruptions to transportation (e.g., railway cancellations, road closures); interruptions to utilities (e.g., power, water supply); failures in communication networks; and industrial facility damage. Both immediate physical damage and service disruptions should be considered.” (p.19) |
| <b>Q136</b> | “Agricultural Impact: Focuses on weather-related effects on farming and livestock management. Includes crop yield variations; direct damage to crops, timber, or livestock; modifications to farming schedules; disruptions to food production and supply chains; impacts on farming equipment; and changes in agricultural inputs (e.g., soil conditions, water availability, fertilizers, animal feed). Both immediate and long-term effects should be considered.” (p.19)        |
| <b>Q137</b> | “Ecological Impact: Examines effects on natural environments and ecosystems. Includes changes in biodiversity; impacts on wildlife populations and behavior; effects on non-agricultural plant life; habitat modifications (e.g., forests, wetlands, water bodies); changes in hydrological systems (e.g., river levels, lake conditions); and urban plant life impact. Immediate environmental changes should be prioritized.” (p.19)                                              |
| <b>Q138</b> | “Financial Impact: Analyzes economic consequences of weather events. Includes direct monetary losses; business disruptions requiring financial intervention; market fluctuations; impacts on tourism and local economies; and insurance claims or economic relief measures. The focus should be on explicit financial impacts rather than inferred consequences.” (p.19)                                                                                                            |

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| <b>Q139</b> | "Human Health Impact: Examines both physical and mental health effects. Includes direct injuries or fatalities (including cases where one or more casualties are explicitly mentioned); increased risks of weather-related illnesses; mental health consequences (e.g., stress, anxiety); impacts on healthcare accessibility; and long-term health implications. Both short-term and chronic health effects should be considered." (p.19)                                                                                                                                                     |
| <b>Q140</b> | "Political Impact: Evaluates governmental and policy responses to weather events. Includes government decision-making and policy changes; shifts in public opinion or political discourse; effects on electoral processes; international aid and relations; and debates on disaster preparedness and response. Both direct political reactions and policy implications should be analyzed." (p.19)                                                                                                                                                                                             |
| <b>Q141</b> | "Infrastructural Impact: Classify as 'true' if the text mentions any damage or disruption to physical infrastructure and essential services. This includes structural damage to buildings, roads, or bridges; any disruptions to transportation systems such as railway cancellations or road closures; interruptions to public utilities including power and water supply; any failures in communication networks; or damage to industrial facilities. Consider only explicit mentions of physical damage or service disruptions in your classification." (p.20)                              |
| <b>Q142</b> | "Agricultural Impact: Classify as 'true' if the text mentions any weather-related effects on farming and livestock management operations. This includes yield variations in crops and animal products; direct damage to crops, timber resources, or livestock; modifications to agricultural practices or schedules; disruptions to food production or supply chains; impacts on farming equipment and resources; or effects on agricultural inputs including soil conditions, water availability for farming, and essential materials such as seedlings, fertilizers, or animal feed." (p.20) |
| <b>Q143</b> | "Ecological Impact: Classify as 'true' if the text mentions any effects on natural environments and ecosystems. This includes alterations to local environments and biodiversity; impacts on wildlife populations and behavior patterns; effects on non-agricultural plant life and vegetation; modifications to natural habitats including water bodies, forests, and wetlands; changes in hydrological systems such as river levels and lake conditions; or impacts on urban plant life." (p.20)                                                                                             |
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| <b>Q145</b> | "Human Health Impact: Classify as 'true' if the text mentions physical or mental health effects of weather events on populations. This includes direct injuries or fatalities (including cases where zero or more casualties are explicitly mentioned); elevated risks of weather-related or secondary illnesses; mental health consequences such as stress or anxiety; impacts on healthcare service accessibility; or long-term health implications." (p.20)                                                                                                                                 |
| <b>Q146</b> | "Political Impact: Classify as 'true' if the text mentions governmental and policy responses to weather events. This includes government decision-making and policy modifications in response to events; changes in public opinion or political discourse; effects on electoral processes or outcomes; international relations and aid responses; or debates surrounding disaster preparedness and response capabilities." (p.20)                                                                                                                                                              |
| <b>D10</b>  | Residents of the locality were put to great inconvenience by the huge piles of snow now accumulated on the streets... Already \$30,000 has been expended in removing it — 1887 Montreal snow removal municipal debate — multi-label (infra+pol+fin) correctly labeled                                                                                                                                                                                                                                                                                                                          |

## Checklist

### Checklist item definitions:

| ID          | Assessment Question                                          |
|-------------|--------------------------------------------------------------|
| <b>OO-1</b> | Review output label categories for regional relevance        |
| <b>OO-2</b> | Identify missing regionally specific categories              |
| <b>OO-3</b> | Flag categories encoding non-regional values                 |
| <b>OO-4</b> | Consider stakeholder-driven taxonomy redesign                |
| <b>OO-5</b> | Document taxonomy issues harming structural/content validity |

### Checklist responses:

**OO-1:** Six labels nominally cover broad disaster consequences but operationalize them via Canadian exemplars [\[Q141–Q146\]](#); boundary logic does not match Gran Chaco consequence-framing [\[regional\\_consequence\\_framing\\_dependency\]](#).

**OO-2:** Missing categories: cross-border displacement (regional-priority, ABSENT), border integrity (ABSENT), indigenous-community-access disruption, strategic economic route disruption, locust outbreaks. Confirmed by [\[WEB-5, WEB-1\]](#) and dataset gap [\[D10, D18\]](#).

**OO-3:** Political category encodes North American policy/governance assumptions (municipal budget debates, electoral processes) [Q140, D10]; Ecological category centers North American ecosystems (Everglades [D19]); neither encodes Southern Cone or border-region values.

**OO-4:** Stakeholder-driven taxonomy redesign would be required: AFE/SINAGIR, SEN, and VIDEI national coordinators would need to define operational labels reflecting consequence-framing logic [WEB-16, WEB-20, WEB-22].

**OO-5:** Structural validity is compromised by the operational definitions; content validity is compromised by missing categories; external validity is compromised by definitional drift between Canadian and Gran Chaco framings.

| ID   | Evidence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| Q25  | "Six vulnerability-related disruptive weather impacts are defined as the labeling categories, including Infrastructural, Political, Financial, Ecological, Agricultural, and Human Health." (p.4)                                                                                                                                                                                                                                                                                                                                                                                              |
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| Q117 | "The classification task is binary (true/false), requiring the model to identify whether the text explicitly mentions any of the defined impacts." (p.16)                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Q140 | "Political Impact: Evaluates governmental and policy responses to weather events. Includes government decision-making and policy changes; shifts in public opinion or political discourse; effects on electoral processes; international aid and relations; and debates on disaster preparedness and response. Both direct political reactions and policy implications should be analyzed." (p.19)                                                                                                                                                                                             |
| Q141 | "Infrastructural Impact: Classify as 'true' if the text mentions any damage or disruption to physical infrastructure and essential services. This includes structural damage to buildings, roads, or bridges; any disruptions to transportation systems such as railway cancellations or road closures; interruptions to public utilities including power and water supply; any failures in communication networks; or damage to industrial facilities. Consider only explicit mentions of physical damage or service disruptions in your classification." (p.20)                              |
| Q142 | "Agricultural Impact: Classify as 'true' if the text mentions any weather-related effects on farming and livestock management operations. This includes yield variations in crops and animal products; direct damage to crops, timber resources, or livestock; modifications to agricultural practices or schedules; disruptions to food production or supply chains; impacts on farming equipment and resources; or effects on agricultural inputs including soil conditions, water availability for farming, and essential materials such as seedlings, fertilizers, or animal feed." (p.20) |
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| Q146 | "Political Impact: Classify as 'true' if the text mentions governmental and policy responses to weather events. This includes government decision-making and policy modifications in response to events; changes in public opinion or political discourse; effects on electoral processes or outcomes; international relations and aid responses; or debates surrounding disaster preparedness and response capabilities." (p.20)                                                                                                                                                              |
| Q147 | "Return 'false' for any impact category that is either not present in the text or not related to weather events. Base classifications on explicit mentions in the text. Focus on direct impacts rather than implications. Consider immediate and direct effects." (p.20)                                                                                                                                                                                                                                                                                                                       |
| D10  | Residents of the locality were put to great inconvenience by the huge piles of snow now accumulated on the streets... Already \$30,000 has been expended in removing it — 1887 Montreal snow removal municipal debate — multi-label (infra+pol+fin) correctly labeled                                                                                                                                                                                                                                                                                                                          |
| D16  | More than 292,000 people have been evacuated in the mountainous region along the Yangtze River, with more than 100,000 homes damaged and crops on about 175,000 hectares of farmland destroyed — 2007 European/Chinese/Pakistani multi-location disaster report from Montreal Gazette — agricultural and health labeled, but not infrastructural despite 100k homes damaged                                                                                                                                                                                                                    |
| D18  | A line of freak thunderstorms rumbling across the northeastern Mediterranean coast from Spain and southern France to the principality of Monaco brought two days of disastrous flash flooding... Raging waters uprooted trees and swept away cars as river levels rose by as much as 3 metres — 1992 Mediterranean flash flood — political and ecological but not infrastructural ("swept away cars," "uprooted trees") — label boundary question                                                                                                                                              |

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| D19    | With about half the original 1.6-million-hectare swamp filled for development or drained for agriculture, the park includes about 202,000 hectares of marsh experts liken the Everglades to a seriously ill patient — 1990 Everglades drought/ecosystem article — ecological only; US content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| WEB-1  | Paraguayan emergency Ley N° 7471/2025 frames flood emergency at the departmental level — political framing absent from benchmark exemplars ( <a href="https://www.bacn.gov.py/leyes-paraguayas/12739/ley-n-7471-2025-que-declara-en-situacion-de-emergencia-a-los-departamentos-de-presidente-hayes-boquer-n-y-alto-paraguay-y-ampl-a-el-presupuesto-general-de-la-nacion-para-el-ejercicio-fiscal-2025-aprobado-por-ley-n-7408-de-fecha-30-de-diciembre-de-2024-presidencia-de-la-republica-secretaria-de-emergencia-nacional">https://www.bacn.gov.py/leyes-paraguayas/12739/ley-n-7471-2025-que-declara-en-situacion-de-emergencia-a-los-departamentos-de-presidente-hayes-boquer-n-y-alto-paraguay-y-ampl-a-el-presupuesto-general-de-la-nacion-para-el-ejercicio-fiscal-2025-aprobado-por-ley-n-7408-de-fecha-30-de-diciembre-de-2024-presidencia-de-la-republica-secretaria-de-emergencia-nacional</a> ) |
| WEB-5  | Cross-border displacement and indigenous community evacuation are primary Gran Chaco disaster consequences (10,000+ Wichí/Toba/Chorote/Tapiete evacuated in 2018) ( <a href="https://www.iadb.org/en/staying-ahead-gran-chacos-floods">https://www.iadb.org/en/staying-ahead-gran-chacos-floods</a> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| WEB-16 | AFE/SINAGIR is the current Argentine national authority — designated as authoritative labelers by deployment user, with zero benchmark representation ( <a href="https://www.argentina.gob.ar/sinagir/institucional">https://www.argentina.gob.ar/sinagir/institucional</a> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| WEB-20 | SEN (Paraguay) coordinates emergencies in all three Chaco departments — also unrepresented ( <a href="https://sen.gov.py/institucional/">https://sen.gov.py/institucional/</a> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| WEB-22 | VIDECI (Bolivia) operates departmental representation in Tarija — also unrepresented ( <a href="https://www.mindef.gob.bo/node/499">https://www.mindef.gob.bo/node/499</a> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

## Information Gaps

- Gran Chaco consequence-framing logic has not been formally documented as an annotation guideline

## Requires Expert Verification

- AFE/SINAGIR national coordinators should review the six-category definitions and propose extensions or replacements

## Remediation

**Gap 1: Label boundaries operationalized via Canadian exemplars; consequence-framing-dependent assignment is unsupported.**

**Recommendation:** Rewrite operational definitions ([Q141–Q146]) with Gran Chaco exemplars elicited from AFE/SINAGIR/SEN/VIDECI coordinators; explicitly document consequence-framing decision rules for ambiguous multi-label cases (e.g., flooded dirt road blocking access to indigenous community).

| ID   | Evidence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| Q141 | “Infrastructure Impact: Classify as ‘true’ if the text mentions any damage or disruption to physical infrastructure and essential services. This includes structural damage to buildings, roads, or bridges; any disruptions to transportation systems such as railway cancellations or road closures; interruptions to public utilities including power and water supply; any failures in communication networks; or damage to industrial facilities. Consider only explicit mentions of physical damage or service disruptions in your classification.” (p.20)                               |
| Q142 | “Agricultural Impact: Classify as ‘true’ if the text mentions any weather-related effects on farming and livestock management operations. This includes yield variations in crops and animal products; direct damage to crops, timber resources, or livestock; modifications to agricultural practices or schedules; disruptions to food production or supply chains; impacts on farming equipment and resources; or effects on agricultural inputs including soil conditions, water availability for farming, and essential materials such as seedlings, fertilizers, or animal feed.” (p.20) |
| Q143 | “Ecological Impact: Classify as ‘true’ if the text mentions any effects on natural environments and ecosystems. This includes alterations to local environments and biodiversity; impacts on wildlife populations and behavior patterns; effects on non-agricultural plant life and vegetation; modifications to natural habitats including water bodies, forests, and wetlands; changes in hydrological systems such as river levels and lake conditions; or impacts on urban plant life.” (p.20)                                                                                             |
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| Q145 | “Human Health Impact: Classify as ‘true’ if the text mentions physical or mental health effects of weather events on populations. This includes direct injuries or fatalities (including cases where zero or more casualties are explicitly mentioned); elevated risks of weather-related or secondary illnesses; mental health consequences such as stress or anxiety; impacts on healthcare service accessibility; or long-term health implications.” (p.20)                                                                                                                                 |

**Q146**

"Political Impact: Classify as 'true' if the text mentions governmental and policy responses to weather events. This includes government decision-making and policy modifications in response to events; changes in public opinion or political discourse; effects on electoral processes or outcomes; international relations and aid responses; or debates surrounding disaster preparedness and response capabilities." (p.20)